

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 – 2024

Data Source: Global Carbon Budget 2025, NASA Mauna Loa Observatory

Atmospheric CO₂ Reaches 429 ppm

50% Above Pre-Industrial Levels

429

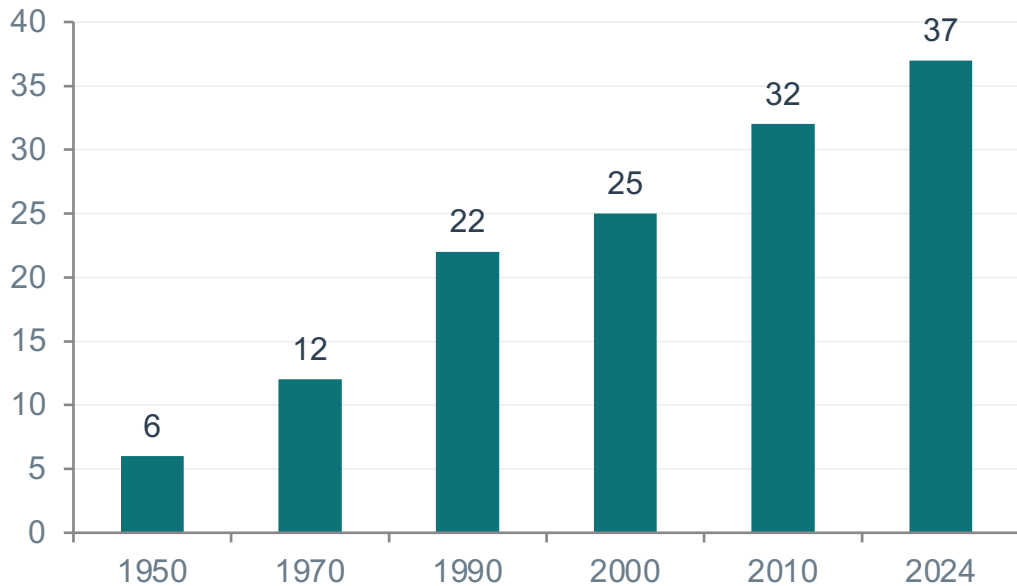
ppm (February 2026)

Mauna Loa Observatory

- Pre-industrial baseline: ~280 ppm
- Current level 1.5× greater than the natural increase at the end of the last ice age
- That natural shift took thousands of years; the current rise occurred in ~2 centuries
- Driven primarily by burning coal, oil, and natural gas

Global Fossil Fuel CO₂ Emissions Grew Six-Fold

6 Billion to 35+ Billion Tonnes



Key Inflection Points

6 Bt — World emissions in 1950

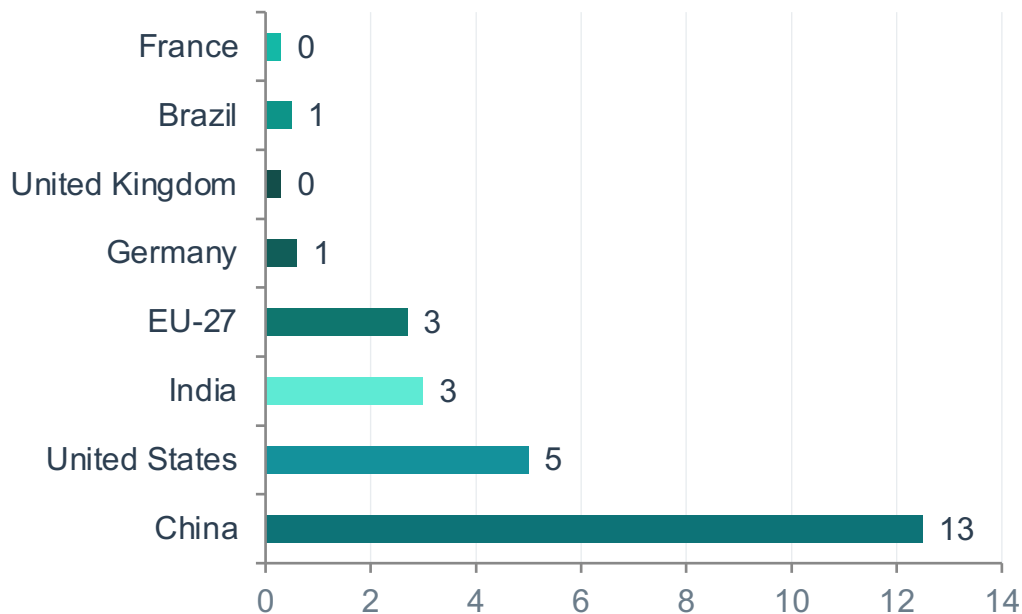
20 Bt — Crossed by 1990

35+ Bt — Current annual emissions

Growth has slowed but not yet peaked

China Now Emits Over 12 Billion Tonnes

More Than Double the United States



Key Insights

China: ~27% of global total

US peaked near 6 Bt around 2005;
now ~5 Bt

India rising rapidly to ~3 Bt

Europe's share has fallen substantially

Asia Now Accounts for ~50% of Global Emissions

US + Europe Below One-Third

1900

Europe + US: >90%

Rest of World: <10%

>90%



2024

Asia ~50%

US+EU <33%

Rest

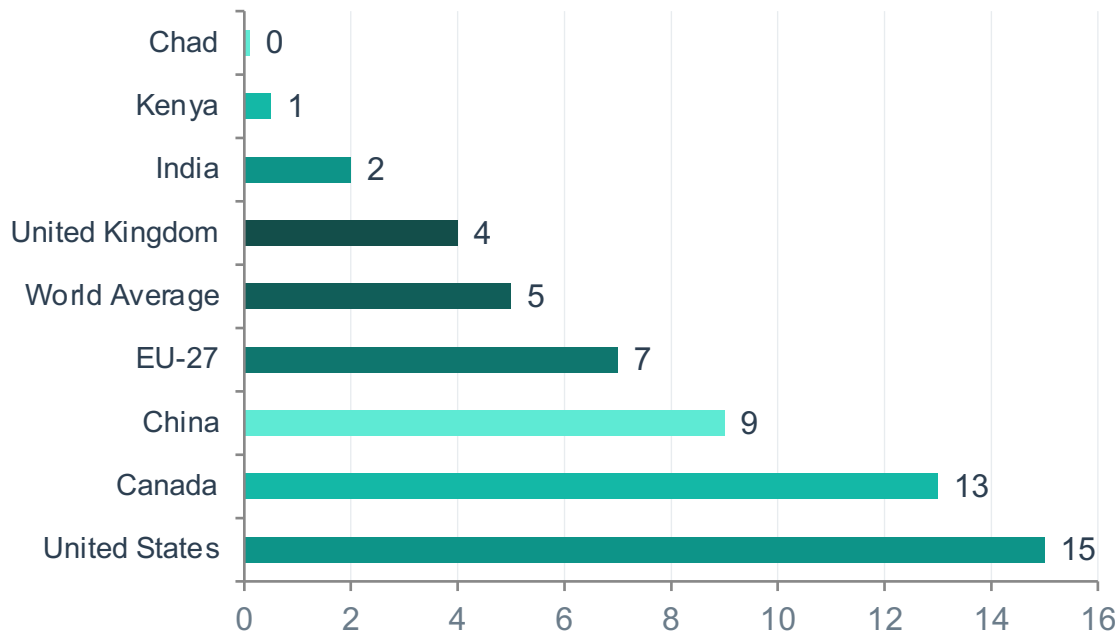
Africa: 3–4%

South America: 3–4%

(Comparable to international aviation)

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad

A 150× Disparity



The Gap

150×

An average American emits in
2 days what an average person in
Chad emits in an
entire year

2024 Snapshot: US and Europe Cut Emissions

While Russia and SE Asia Increase

DECLINES (Blue)

United States: -1% to -5%

Much of Europe: -1% to -5%

Shown in blue on the 2024 map

INCREASES (Orange/Red)

Russia: +2% to +10%

Southeast Asia: +2% to +10%

Several African nations: +2% to +10%+

Some South American countries: >+10%

Note: Some countries show mixed patterns. Percentage changes calculated year-over-year.

Energy Mix Explains Per Capita Differences

Why France Emits Far Less Per Capita Than Germany

	France	United Kingdom	Germany	Netherlands
Primary Energy Sources	Nuclear (70%+), renewables	Nuclear, wind, gas	Coal, renewables, gas	Natural gas, coal
Per Capita CO ₂ (t/person)	~4	~4	~8	~7
Living Standard	High (GDP/capita similar to Germany)	High	High	High

Key Insight

Countries with similar living standards can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels. Policy and infrastructure choices — not just prosperity — determine emissions outcomes.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- 1 Global CO₂ emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
- 2 China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.
- 3 The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third.
- 4 Per capita inequalities are extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
- 5 Energy mix and policy choices matter: countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.